Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

For layered thin films, the central structural question is often not only which phase is present, but how the crystallites are oriented relative to the substrate. In layered van der Waals solids and related two-dimensional materials, this orientational state can directly control the properties that motivate the films in the first place.[? ? ?] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[?] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[?] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[?] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[? ? ? ? ? ? ? ? ?]

Wide-angle x-ray scattering (WAXS) with an area detector is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[? ? ? ? ?] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[?] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an aligned layer normal while remaining azimuthally randomized in-plane.[? ? ?] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[?]

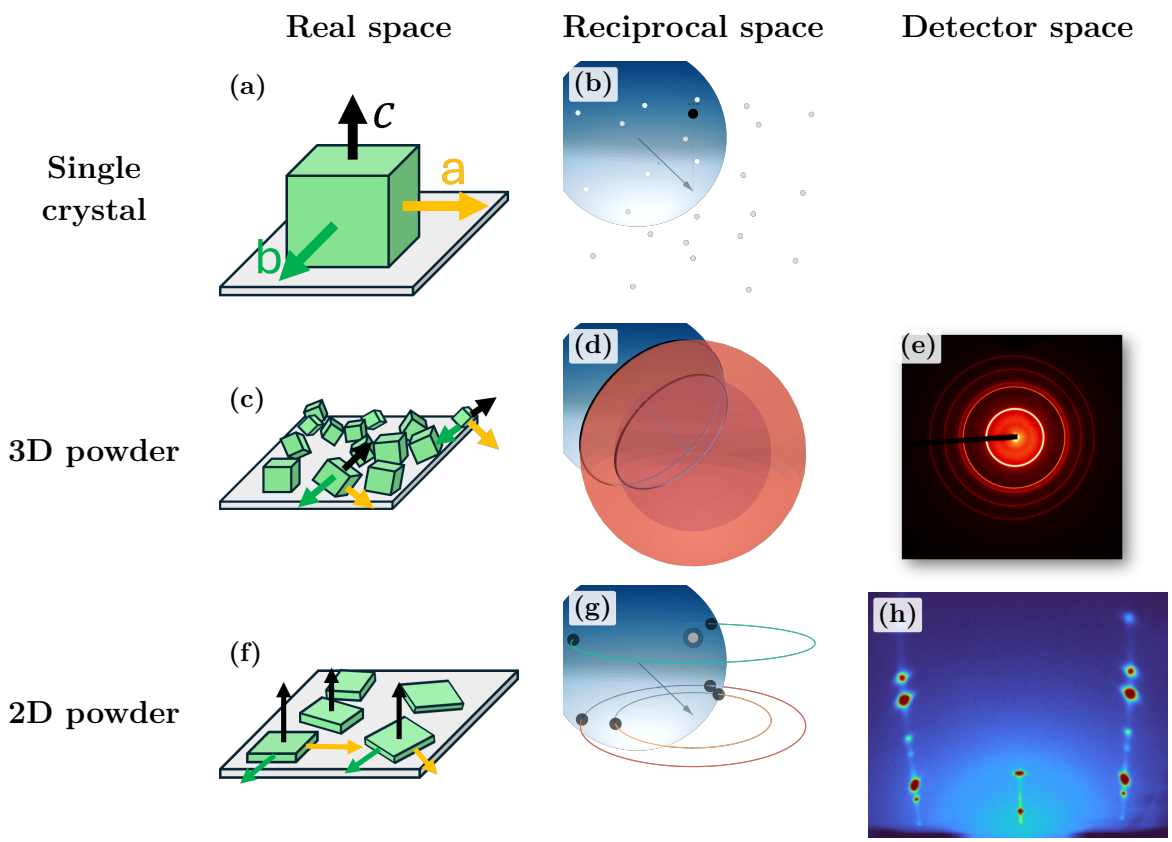

Figure 1: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[?] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[?] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1d, Fig. 1e).[?] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1g).[? ?] The corresponding illustration in Fig. 1f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 1h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near

the specular stripe from $(00L)$ intensity.[? ? ?] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[? ?]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[? ? ? ? ?] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[? ?] Stacking disorder is also commonly inferred from 1D interference models and refinements,[? ? ?] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[? ? ? ? ? ?] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[? ?] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[?] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[? ? ?] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space forward model, using off-specular arcs and specular-family line shapes as direct constraints.[? ?]

Here we present a detector-space forward model for 2D-powder WAXS. Starting from an indexed crystallographic model, it predicts area-detector images and applies the same projection and integration procedures to measured and calculated intensity. The comparison uses detector-space peak positions, local line shapes, relative Bragg intensities, and projected Q_z profiles. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[? ?] reciprocal-space mapping and indexing for oriented thin films,[? ? ? ?] and forward calculations of diffraction from textured films.[?] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing reciprocal-space diffraction signatures, including Nookiin.[?] In contrast, the present work carries the calculation in detector space through detector geometry, grazing-incidence propagation, beam phase space, mosaicity, resolution, and ordered structure-factor weighting. The main-text validation is centered on the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films.[? ?] After the ordered-film validation, we assemble PbI_2 as a diffuse-scattering extension in which stacking disorder changes the structure factor along fixed-Q_R rods while the same detector-space projection and mosaicity framework are retained.

2 Diffraction Model

The detector-space forward model redistributes the structure-factor intensity of the contributing reflections over mosaic distributions. We Monte Carlo sample the incident beam profile across those orientations, weights each ray by transmission, and maps the summed intensity to detector pixels. The calculated and measured images are then compared using the same peak positions, line shapes, relative Bragg intensities, and projected L-integrated profiles.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$, and the simplest diffracted ray is obtained from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

For each trial incident ray, the model computes the sample intersection, applies the incidence geometry and optical corrections, finds the reciprocal-space intersections that satisfy elastic scattering, and propagates the resulting exit rays to the detector. Rays are Monte Carlo sampled from the probability distribution formed by the shape, bandwidth, and divergence of the beam. From every Bragg peak's structure factor spread upon the mosaic, a second Monte Carlo sampling occurs for each ray to choose an intersection point. Footprint, detector tilt, refraction, and absorption enter as experimental resolution and intensity terms. The coordinate conventions for the lab frame, detector plane, sample rotation, and illuminated footprint are summarized in Figs. 2 and 3.

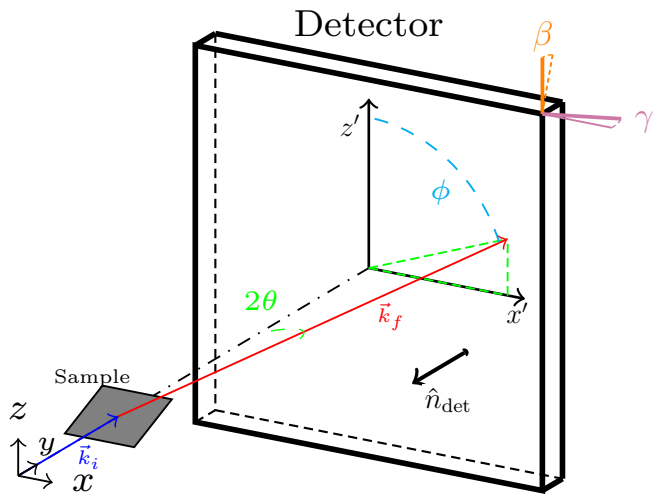

Figure 2: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

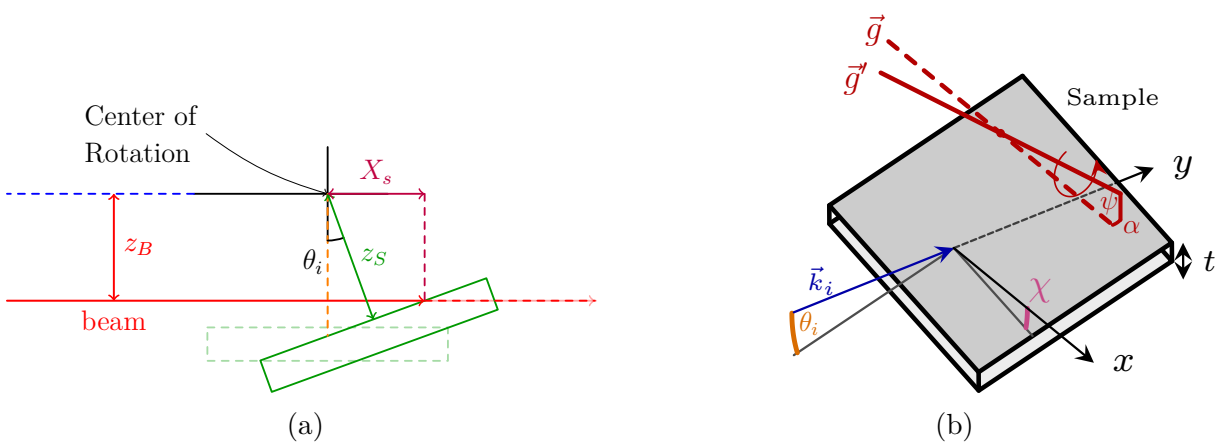

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate integer in-plane Miller pairs (H, K), construct the reciprocal-space rod along G_z for each pair, and evaluate its detector intersections. For labels and profile extraction,

the in-plane pair is represented by the scalar family index m, defined by the in-plane sum

$$m(H,K) = \begin{bmatrix} H & K \end{bmatrix} \mathbf{M}_{\parallel} \begin{bmatrix} H \\ K \end{bmatrix}.$$

Here $\mathbf{M}_{\parallel}$ is the in-plane reciprocal-lattice metric, with any lattice-dependent normalization absorbed into the definition of m. For the hexagonal bismuth chalcogenides, this scalar is proportional to $H^2 + H\,K + K^2$. Thus, m labels an in-plane reciprocal-space rod family. In 2D-powder calculations, all integer (H,K) rods with the same m, and therefore the same Q_r, are evaluated individually and then summed after projection. Multiplicity and orientation effects therefore emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a\, f_a(Q)\, \exp[2\pi i\, \mathbf{G}\cdot\mathbf{r}_a]\, \exp\Big[-2\pi^2\, \mathbf{G}\cdot\mathbf{U}_a\cdot\mathbf{G}\Big], \tag{1}$$

where the sum runs over atoms a in the unit cell. Here o_a, $f_a(Q)$, $\mathbf{r}_a$, and $\mathbf{U}_a$ are the occupancy, atomic form factor, fractional position, and anisotropic displacement tensor for atom a. Symmetry constraints are retained during refinement, so only the independent crystallographic parameters are varied. The detector weight for each allowed ray is multiplied by $|F_{\mathbf{G}}|^2$ before mosaic and experimental broadening are applied.

2.3 Mosaicity

Mosaicity is the out-of-plane orientational spread of the layered crystallites about the mean film normal. It is distinct from the random in-plane rotation that defines the two-dimensional powder limit. Random in-plane rotation generates the equivalent m-indexed rod family at a common in-plane reciprocal-space radius Q_R, while mosaicity determines which tilted members of that family can satisfy the Ewald condition at a fixed incident angle.

The reason mosaicity must be introduced early is visible in the detector data themselves. A single incident angle should select only a limited part of the specular $m = 0$ family if the film has only a narrow orientational distribution. The measured images instead contain unexpected $00L$ intensity, including a low-L feature near the 003 region and weak off-condition specular peaks that remain visible even when the strongest peaks are plotted on a logarithmic scale. Figure 4 should show the full measured detector image before any model overlay, and Fig. 5 isolates the local feature that motivates the long-tail component of the mosaic model.

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`figures/mosaic/raw_detector_mosaic_motivation.png`

Full raw or minimally processed detector image. Show the measured diffraction pattern at the incident angle used to introduce the mosaicity problem; include a crop box or unobtrusive marker indicating the low-L $m = 0$ region shown in the next figure.

Figure 4: Raw detector image used to motivate the mosaicity model. This figure should precede the schematic treatment so the reader first sees the experimental problem: a single incident-angle image contains off-specular rod-family arcs, strong specular-family peaks, and low-L $m = 0$ intensity that cannot be understood from a narrowly aligned film alone. The plotted image should be measured data, not a simulation, with minimal annotation beyond the region selected for the following close-up.

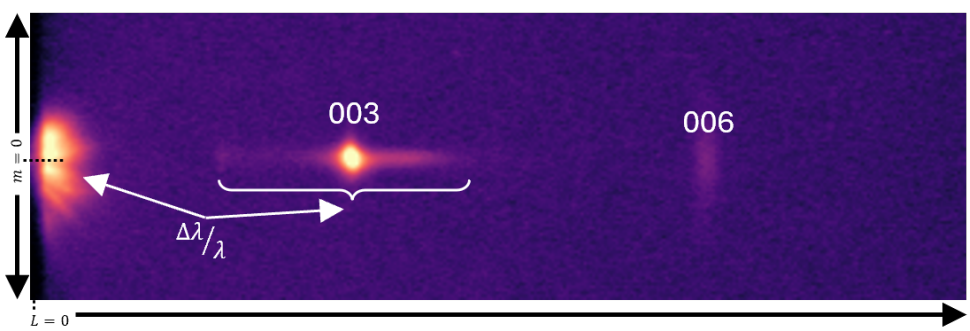

Figure 5: Close-up of the low-L $m = 0$ detector feature. The bright near-origin intensity and vertical star-like morphology are not expected from an ideal single-crystallite calculation or from a mosaic distribution with only a rapidly decaying core. The feature reflects the combined effect of wavelength bandwidth and angular divergence, which give the Ewald sphere an effective thickness, and a long mosaic tail, which supplies tilted crystallites that bring otherwise off-condition specular-family intensity into the measured detector region.

The main point is not simply that the peaks are broad. The in-plane rod-family arcs are largely controlled by the narrow mosaic core and by instrumental resolution. The specular-family features expose a different part of the orientational distribution: weak intensity from crystallites tilted far enough from the mean normal to satisfy reflections that would otherwise be missed at the nominal incidence condition. A rapidly decaying Gaussian-like mosaic

density can reproduce the central widths of near-condition features but does not provide enough tilted-crystallite population in the tails. We therefore use a compact two-component angular density,

$$\omega_{\mathrm{w}}(\Delta\theta;\eta,\Gamma,\sigma) = \eta L_{\mathrm{w}}(\Delta\theta;\Gamma) + (1-\eta)G_{\mathrm{w}}(\Delta\theta;\sigma), \qquad 0 \le \eta \le 1. \tag{2}$$

Here G_{w} is the Gaussian-like core, L_{w} is the Lorentzian-like tail, η is the Lorentzian weight, and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal. The two widths are independent: the Lorentzian full width at half maximum is 2Γ, while the Gaussian full width at half maximum is $2\sqrt{2\ln 2}\,\sigma$. The wrapped form conserves probability over one angular period and avoids an artificial discontinuity at $\Delta\theta = \pm\pi$.

The Lorentzian form should be read as a useful long-tail approximation rather than as a claim that the microscopic distribution is exactly Lorentzian. What matters experimentally is the separation of roles. The Gaussian-like core controls the dominant widths of the strong, near-condition features, while the long tail accounts for weak specular-family intensity that appears outside the narrow-core expectation. This is why the mosaic model is constrained most directly by the $m = 0$ profiles and by the low-L detector feature, not only by the widths of the brighter off-specular arcs.

Figure 6 summarizes how the same orientational distribution appears in reciprocal space and detector space. For $G_R \neq 0$, random in-plane rotation and out-of-plane mosaicity produce off-specular arcs. For $G_R = 0$, the same angular distribution produces cap-like $00L$ intensity near the specular direction, where the tail contribution is especially visible.

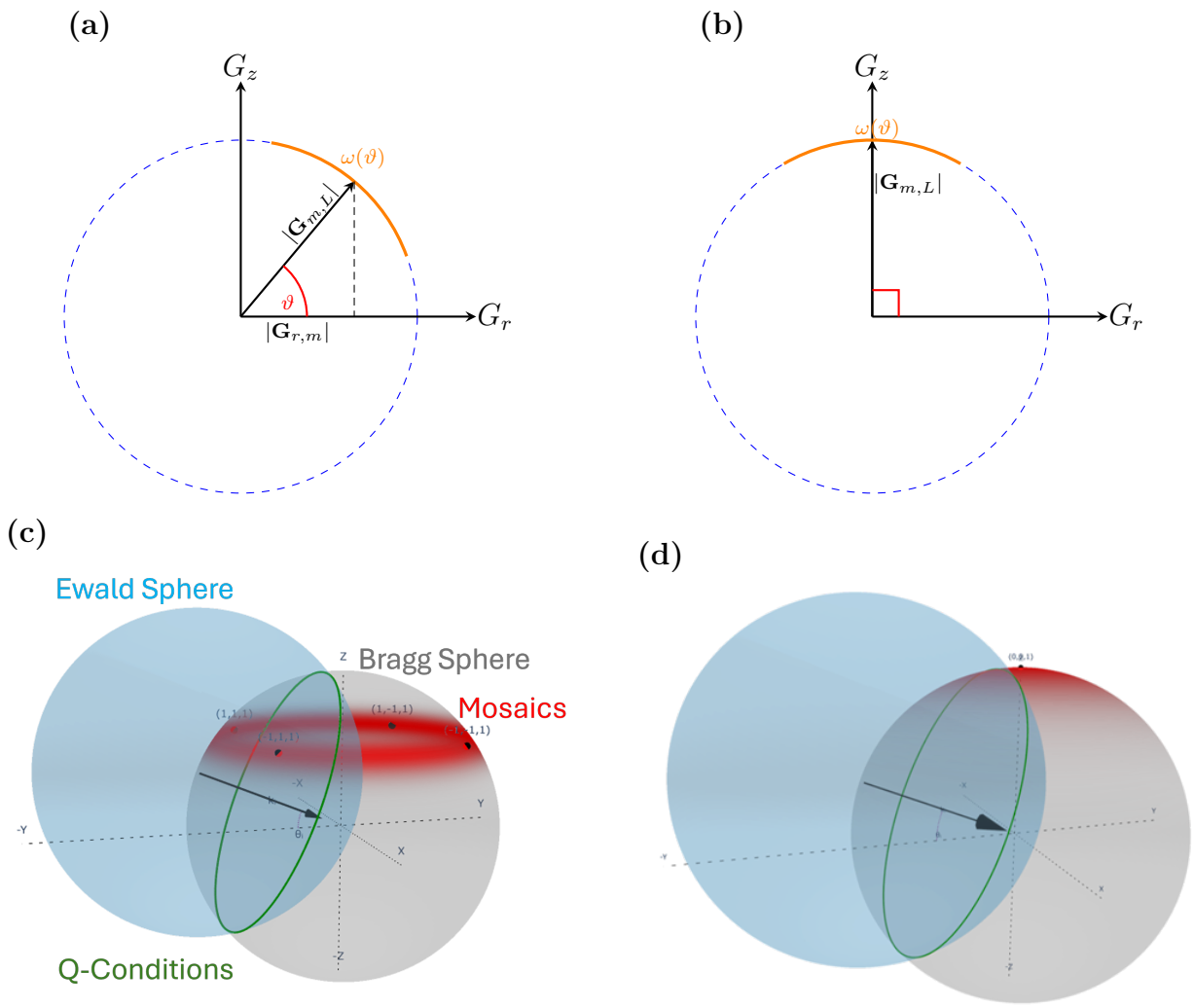

Figure 6: Mosaic broadening in reciprocal space and detector space. (a,c) For $G_R \neq 0$, random in-plane rotation and finite out-of-plane mosaicity give off-specular arcs or bands. (b,d) For $G_R = 0$, the same orientational distribution gives cap-like $00L$ intensity near the specular direction. The narrow Gaussian-like core controls the main width of near-condition features, while the Lorentzian-like tail supplies the weak tilted-crystallite population needed for off-condition $m = 0$ peaks.

The next section treats the coupled bandwidth, divergence, and Bragg-sphere-size effects that determine which part of this mosaic distribution is actually sampled by the detector.

2.4 Correlated mosaicity, bandwidth, and Bragg-sphere size

Mosaicity is not observed independently of the beam phase space. It is observed through the subset of mosaic orientations that also satisfy the Ewald condition for the incident wavelength, incident direction, and detector geometry. This is why the low-L $m = 0$ features

in Fig. 5 are discussed separately from the mosaic functional form. They depend on the correlated action of mosaicity, wavelength bandwidth, beam divergence, and Bragg-sphere size.

For one monochromatic incident ray, elastic scattering selects the intersection of two geometric objects: the Ewald sphere, with radius $2\pi/\lambda$, and the Bragg sphere associated with a reflection of magnitude $|\mathbf{G}|$. The Bragg sphere is a useful construction because each point on it corresponds to a different crystallite orientation for the same reciprocal-lattice-vector magnitude. Mosaicity populates only a cap-like region of that Bragg sphere around the nominal orientation. The observed intensity is therefore not the whole Ewald–Bragg intersection ring; it is the part of that allowed ring that overlaps the populated mosaic cap.

Finite wavelength bandwidth changes the Ewald sphere from an infinitesimally thin surface into a shell of allowed radii. Beam divergence and incident-angle spread have a similar effect by moving the Ewald-sphere center and direction. When this finite Ewald shell intersects the mosaic-populated Bragg sphere, the overlap region can become much larger than the ideal single-ray intersection. This is especially important for low-order reflections. For a small Bragg sphere, a fixed instrumental bandwidth or angular spread covers a comparatively large fraction of the Bragg sphere and can continue to overlap the mosaic cap over a broad range of nominal incident angles. For a larger Bragg sphere, the same bandwidth produces a narrower effective intersection, so the detected intensity is more sensitive to the exact incidence condition.

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`figures/mosaic/bragg_sphere_bandwidth_size_series.png`

Tutorial schematic series showing a finite-thickness Ewald sphere intersecting small, medium, and large Bragg spheres. Include the mosaic cap on each Bragg sphere and highlight how much of the cap is sampled for each sphere size.

Figure 7: Planned tutorial schematic for the coupled bandwidth–mosaicity geometry. The figure should compare small and large Bragg spheres, show the finite Ewald-sphere thickness from wavelength bandwidth, overlay the mosaic cap, and identify the allowed intersection ring. The central lesson is that the same instrumental bandwidth can sample a much larger angular fraction of a low-order Bragg sphere than of a high-order Bragg sphere.

This geometry explains a counterintuitive feature of the measurements. Increasing the incidence angle can bring high-order $00L$ peaks into condition while the low-order peaks remain visible, but lowering the incidence angle does not necessarily preserve the high-order peaks. The low-order peaks are robust because the finite Ewald-shell thickness and the mosaic cap continue to overlap. The high-order peaks require a more specific intersection because their Bragg spheres are larger and the same bandwidth samples a smaller fraction of the populated cap.

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`figures/mosaic/delta_lambda_detector_series.png`

Calculated detector-image sequence for different $\Delta\lambda$ values and/or incident angles. Show how the low-L feature, streaks, and star-like morphology change as the Ewald-shell thickness changes.

Figure 8: Planned calculated-image sequence connecting the schematic geometry to detector space. The panels should hold the mosaic model fixed while varying $\Delta\lambda$, incident angle, or Bragg-sphere size, then show how the detector streaks and star-like $m = 0$ morphology change. This figure is intended to make the correlated effects visible rather than leaving them as text-only geometry.

The detector image is the projection of the allowed $\mathbf{k}_f$ directions. The Ewald–Bragg intersection ring is a geometric set of possible rays, not a statement that intensity is uniformly present around the ring. Most of the ring is dark unless it crosses the mosaic-populated cap. When it does cross the cap, the mapped exit vectors can form a narrow streak, a star-like feature, or broadened cap-like intensity depending on $|\mathbf{G}|$, incident angle, bandwidth, divergence, and the tail of ω_{w}. The low-L feature is therefore not only evidence for a long mosaic tail. The long tail supplies tilted crystallites, but finite bandwidth and divergence determine which part of that tail the detector samples.

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`figures/mosaic/low_q_003_03_06_data_model_overlay.png`

Side-by-side measured and calculated detector crops/profiles for the low-Q or 003 region, the 03 feature, and the off-condition 06 or 006 feature. Include a with/without bandwidth or with/without Lorentzian-tail comparison if available.

Figure 9: Planned data/model comparison for the correlated-effects explanation. The figure should place the measured and calculated low-Q or 003 feature, the 03 feature, and the off-condition 06/006 feature next to one another. A useful final version would include a control calculation with the Lorentzian tail or finite bandwidth disabled, provided that the control calculation has been verified.

In the final ordered-film validation, the off-specular arcs, the weak off-condition $m = 0$ peaks, and the low-L feature are interpreted together. Off-specular line widths constrain the narrow mosaic core and detector resolution. Weak $00L$ peaks constrain the Lorentzian-like tail. The star-like low-L feature tests whether that mosaic tail has been combined with the correct wavelength bandwidth, divergence, and detector projection. Treating these effects as correlated is what allows the model to reproduce non-obvious detector features rather than only the strongest Bragg peaks.

3 Results: ordered Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films validate the model by showing that the same detector-space forward model quantitatively reproduces measured line shapes and intensities in two related layered chalcogenides. Three detector images, collected at incident angles of 5°, 10°, and 15°, were used in the refinement. The 5° condition gives the clearest single-view display of both the off-specular arc families and the specular $m = 0$ features. Together, the data test the two-dimensional powder orientational limit, sample geometry, finite beam phase space, mosaicity, detector resolution, and ordered structure-factor weighting.

The ordered-film comparison is intentionally made after the mosaic and correlated-effects discussion. The model is not only asked to place intense Bragg features at the correct detector

positions. It must also reproduce weak off-condition $m = 0$ peaks, semilogarithmic tails, and the low-L specular-family intensity that motivate the two-component mosaicity model. A narrow-core mosaic distribution can reproduce the main width of near-condition features, but the measured $m = 0$ profiles show additional $(00L)$ intensity that persists far from the strongest specular peak. Those weak peaks provide the direct experimental check on the Lorentzian-like tail.

The low-L detector feature introduced in Fig. 5 is a second, independent check. It lies close to the direct-beam and critical-angle region, where it could easily be mistaken for ordinary reflectivity or an unrelated detector artifact. The correlated-effects discussion in Sec. 2.4 instead predicts that a small Bragg sphere, finite wavelength bandwidth, finite divergence, and a long mosaic tail can redistribute $m = 0$ intensity into a star-like detector morphology. The ordered-film comparison below therefore has to reproduce more than the off-specular arc positions: it must also account for the unexpected specular-family peaks and the low-L morphology that motivated the mosaic model.

The detector-space panels and profile panels in Fig. 10 use the same extraction convention. In panels (a) and (b), the solid colored curves are the calculated centerlines of selected m-indexed Bragg-rod trajectories on the detector. The semi-transparent colored bands are the finite detector regions integrated to form the one-dimensional profiles. The dashed colored outlines mark the edges of those integration bands. The central blue band is the $m = 0$ specular-family trajectory, while the paired off-specular bands are the two detector-side intersections of the same two-dimensional powder rod family. The plus and minus labels identify those two detector-side branches only.

Panels (c) and (d) show the corresponding line profiles after applying the same projection and integration operation to the measured and calculated images. The horizontal coordinate is L, so a marker labeled $L = n$ denotes the nominal $00n$ or corresponding off-specular Bragg condition for that selected rod family. The black solid curves are the measured profiles. The orange dashed curves are the simulated profiles after the same detector-space projection. The off-specular rows are plotted on a linear intensity scale, while the $m = 0$ row is plotted on a logarithmic scale because the important result is the persistence of weak, unexpected $00L$ intensity far from the strongest specular peak.

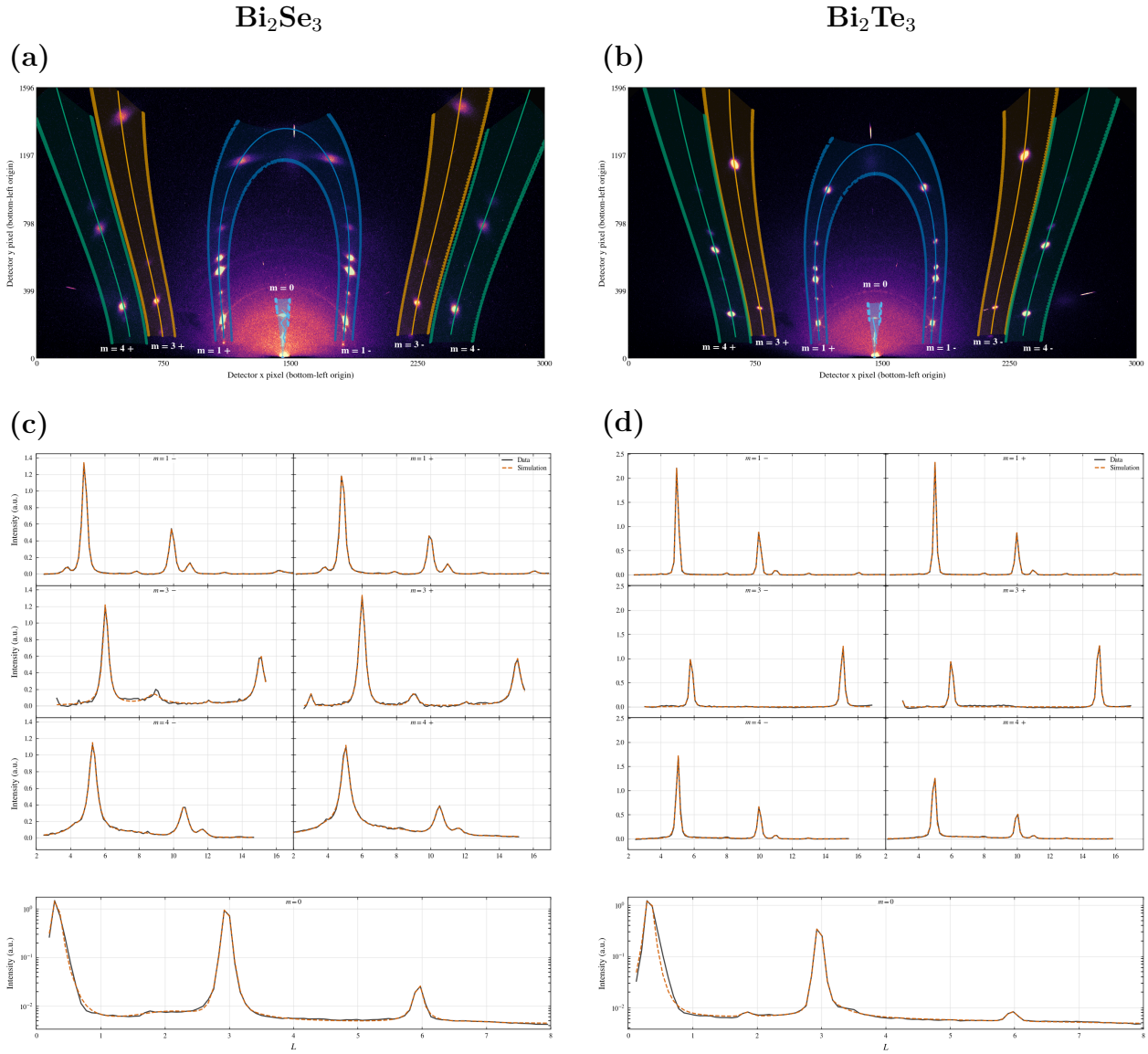

Figure 10: Ordered-film validation of Bi_2Se_3 and Bi_2Te_3 at $\theta_i = 5°$. (a,b) Measured detector images with selected reciprocal-space extraction regions overlaid. Solid colored curves are the trajectory centerlines. Semi-transparent colored regions are the detector pixels integrated to form the profiles. Dashed colored outlines mark the integration-band boundaries. The labels m identify two-dimensional powder rod families, and the signs label the two detector-side branches of the same family. The central $m = 0$ band follows the specular family and includes the bright low-L intensity near the critical-angle region. (c,d) Extracted profiles plotted versus L after applying the same projection to measured and calculated images. Black solid curves are measured data and orange dashed curves are simulations. The off-specular rows test detector geometry, the two-dimensional powder construction, the mosaic-core width, and the ordered structure-factor pattern. The semilogarithmic $m = 0$ rows emphasize the unexpected persistence of additional $00L$ peaks, which requires a long Lorentzian mosaic tail in addition to the narrow mosaic core.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains meaningful because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. The figure therefore tests the detector-space calculation as used experimentally. This distinction is especially important for the unexpected $00L$ peaks and the enhanced low-L specular intensity, where incidence angle, wavelength bandwidth, divergence, detector-coordinate uncertainty, background, and the Lorentzian mosaic tail are most strongly coupled.

In Fig. 10, the off-specular and $m = 0$ profiles play complementary roles. The off-specular arcs check the two-dimensional powder reciprocal-space construction and calibrated detector geometry, while their widths and asymmetries constrain the Gaussian-like mosaic core and instrumental resolution. The off-condition $m = 0$ peaks test the Lorentzian mosaic tail and show why a rapidly decaying mosaic core alone is insufficient. The ordered-film result is therefore not only that the fitted curves match the strongest peaks, but that the same physical model also accounts for weaker and less intuitive detector features that were not built into a one-dimensional reduction.

4 Results: diffuse scattering in PbI_2

PbI_2 extends the ordered-film analysis to a case where the detector image is not described by Bragg peaks from a single ordered stacking sequence alone. The same calibrated detector geometry, beam phase space, two-dimensional powder construction, and mosaic treatment are still used. The additional physics is stacking disorder: local PbI_2 layers remain crystalline enough to define in-plane reciprocal-space radii, but the registry between adjacent layers is not perfectly periodic. That loss of stacking coherence redistributes intensity along Q_z and produces diffuse features that must be modeled rather than subtracted as background.

Draft figure placeholder

`figures/results_pbi2/pbi2_raw_detector_diffuse_motivation.png`

Raw or minimally processed PbI_2 detector image. Mark the diffuse features that motivate the stacking-disorder model, but keep the measured data visually dominant.

Figure 11: Planned PbI_2 detector image motivating the diffuse-scattering section. The figure should show the measured detector image before a detailed model overlay and identify the diffuse intensity associated with stacking disorder. The purpose is to make clear that the PbI_2 section introduces a new physical ingredient beyond the ordered Bi_2Se_3 and Bi_2Te_3 fits: layer-stacking disorder creates intensity distributed along reciprocal-space trajectories rather than only at sharp ordered Bragg positions.

In an ideal layered crystal, the scattering from successive layers has a fixed phase relation. For a reflection with momentum transfer $\mathbf{Q} = (\mathbf{Q}_R, Q_z)$, the intensity contains an interference factor from all layer pairs. If the separation and in-plane registry of every adjacent layer are perfectly repeated, that interference factor is sharply peaked at the ordered Bragg values of Q_z. A stacking fault changes the registry of one layer relative to the next. The in-plane order within each layer can remain largely intact, so the same Q_R families are still present, but the phase relation along the stacking direction is interrupted. The result is a combination of residual ordered Bragg intensity and diffuse intensity spread along Q_z.

For a two-dimensional powder film, random in-plane rotation turns each in-plane reflection family into a reciprocal-space cylinder at fixed Q_R. Stacking disorder then modulates the intensity along that cylinder. In detector space, the measured diffuse features are therefore not arbitrary streaks; they are the detector projection of fault-broadened intensity on these Q_R cylinders after the same beam bandwidth, divergence, mosaicity, detector tilt, and integration effects used for the ordered films are applied.

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`figures/results_pbi2/pbi2_diffuse_cylinder_schematic.png`

Reciprocal-space schematic for PbI_2. Show fixed-Q_R cylinders, ordered Bragg points along Q_z, and diffuse intensity redistributed along the cylinders by stacking disorder.

Figure 12: Planned reciprocal-space schematic for PbI_2 diffuse scattering. Each in-plane reflection family produces a fixed-Q_R cylinder in the two-dimensional powder average. Perfect stacking would place intensity mainly at discrete ordered Bragg positions along the cylinder. Stacking faults reduce the layer-to-layer phase coherence and redistribute part of the intensity along Q_z, producing diffuse features that are then projected onto the detector by the same forward model used for the ordered films.

The stacking-fault model should be introduced as a real-space correlation model rather than as an empirical detector correction. A useful representation is a stack of PbI_2 layers in which each layer-to-layer step can follow the ideal registry or one of the allowed faulted registries. The fault probabilities determine the probability that two layers separated by n spacings remain in the same phase relation. Those pair correlations determine the Q_z dependence of the diffuse intensity. In the limit of zero fault probability, the model returns the ordered Bragg sequence. As the fault probability increases, the ordered peaks lose coherent weight and the diffuse component between or around them grows.

Draft figure placeholder

`figures/results_pbi2/pbi2_stacking_fault_model.png`

Real-space stacking schematic. Show ideal PbI_2 layer sequence, one or more faulted layer registries, the corresponding layer-pair correlations, and the qualitative effect on Q_z intensity.

Figure 13: Planned stacking-fault model schematic. The figure should show how an ideal repeated layer registry differs from a faulted sequence. The important quantity for diffraction is not only whether an individual layer is faulted, but how the fault changes pair correlations along the stacking direction. Those correlations control whether intensity remains concentrated at ordered Bragg positions or is redistributed into diffuse Q_z intensity on the fixed-Q_R cylinders.

The PbI_2 comparison should therefore contain three calculated components: an ordered-baseline calculation, a stacking-disorder calculation, and the measured detector-space or projected Q_z profile to which both are compared. The ordered baseline shows which features would be expected from a perfectly periodic stack. The disorder calculation tests whether the chosen fault probabilities and layer-pair correlations reproduce the diffuse intensity without damaging the detector geometry and mosaicity already established by the ordered-film analysis.

Draft figure placeholder

`figures/results_pbi2/pbi2_diffuse_data_model_overlay.png`

PbI_2 measured/calculated comparison. Include measured intensity, ordered-baseline calculation, stacking-disorder calculation, and residuals for selected fixed-Q_R trajectories or detector regions.

Figure 14: Planned PbI_2 diffuse-scattering validation figure. The comparison should use the same detector-space projection logic as the ordered-film figures: measured and calculated images are projected through matched fixed-Q_R regions and plotted versus Q_z or L. The ordered-baseline calculation should show what is missing without stacking disorder, while the stacking-fault calculation should account for the diffuse intensity associated with fault-broadened layer correlations.

This section is intentionally assembled as a figure-driven scaffold. The final version should keep the PbI_2 result subordinate to the ordered-film validation unless the diffuse-scattering figures are complete enough to support an independent conclusion. The scientific role of PbI_2 is to demonstrate that the same detector-space framework can be extended from ordered Bragg line-shape refinement to disorder-sensitive diffuse scattering when the stacking model is expressed in reciprocal-space terms that map onto the measured detector image.

5 Refinement workflow

The workflow is placed after the ordered and PbI_2 result sections because the reader has now seen what the parameters are meant to explain. The refinement is staged rather than fully global from the beginning. Many parameters affect similar detector-space observables: detector distance shifts peak positions, sample alignment changes the apparent incident condition, mosaicity changes profile widths and off-condition intensity, and structure factors change relative peak heights. A simultaneous fit can therefore hide physical errors by compensating one parameter group with another. The staged workflow fixes the most instrumental quantities first, then introduces sample orientation, mosaicity, ordered intensity, and finally stacking disorder where needed.

The first distinction is between instrumental parameters and sample parameters. Instrumental parameters describe the beam and detector and should be shared across samples collected under the same experimental configuration. Sample parameters describe the film: incidence condition, sample offset, mosaic distribution, structural intensity, and, for PbI_2, stacking disorder. The refinement sequence in Table 1 uses the detector-space observable most sensitive to each group before that group is used in the final image projection.

Table 1: Staged refinement workflow. Steps 1–5 are used for the ordered Bi_2Se_3 and Bi_2Te_3 validation. Step 6 is the added PbI_2 extension for stacking-fault diffuse scattering. All sample images are dark-subtracted before the listed observables are formed, and the final measured/calculated comparison applies the same detector-space projection to data and simulation.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film validation	Selected detector-space Bragg regions and Q_z projections from fixed-m trajectories	Image scale factors and ordered structure-factor parameters	Relative Bragg intensities and measured/calculated line-shape agreement
6	PbI_2 diffuse extension	Fixed-Q_R diffuse regions, ordered-baseline residuals, and projected Q_z profiles	Stacking-fault probabilities or layer-pair correlation parameters	Diffuse intensity along Q_z and separation of ordered and disorder-derived scattering

The instrumental stages fix where an ideal reflection should appear before any sample-specific physics is refined. Direct-beam images define the incident phase space, including beam width and angular divergence. A powder calibrant fixes the detector mapping: distance, beam center, and detector tilts. These two stages are intentionally separated from sample fitting because an incorrect detector geometry can otherwise be absorbed into apparent sample tilt, mosaic broadening, or shifted reciprocal-space trajectories.

The alignment stage uses indexed sample reflections to determine how the film sits in the beam. This stage fixes the nominal incident angle, sample offsets, in-plane rotation, and goniometer misalignment well enough that calculated Bragg-rod trajectories can be overlaid on the detector. Only after this step does it make sense to refine the mosaic distribution, because mosaicity should explain profile shape and off-condition intensity rather than compensate for a misplaced trajectory.

The mosaicity stage constrains the shared Gaussian-plus-Lorentzian angular distribution. Local off-specular profiles constrain the narrow core, while the specular-family tails and low-

L features test the long-tail component. The ordered-film validation then applies the same projection to measured and calculated images and compares profiles along fixed-m trajectories. At that point the structure-factor parameters and image scale factors are constrained by relative intensities, not by detector placement alone.

For PbI_2, the ordered workflow is not discarded. It provides the geometry, projection, and mosaic baseline needed before diffuse scattering can be interpreted. The additional stacking-disorder stage asks whether a physically defined layer-correlation model accounts for the remaining diffuse intensity along fixed-Q_R cylinders. This keeps the PbI_2 result tied to the same detector-space framework while making clear which extra parameters belong specifically to stacking faults.

Implementation details, including lookup tables, Monte Carlo sampling, sub-pixel remapping, parameter-correlation plots, and the propagated uncertainty from 2θ–ϕ and Q transformations, should be documented in the supporting information. The main-text role of the workflow is to make the logic of the refinement reproducible: determine instrumental geometry first, determine sample orientation second, refine mosaic/profile effects third, and only then interpret ordered or diffuse intensity differences as structural information.

6 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 films validate the mosaic model and detector-space forward model. A single calculation accounts for detector geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor weighting, then compares measured and calculated profiles along the same fixed-m trajectories. The key result is direct line-shape agreement in detector-derived projections, including features that would be difficult to interpret from a one-dimensional reduction alone.

Keeping the comparison in detector space is essential because the constraining intensity is distributed among off-specular arcs, cap-like $m = 0$ features, weak off-condition specular peaks, and bright low-L specular-region intensity that a conventional one-dimensional reduction would obscure. Because the calculated image includes the same effects and is projected in the same way as the measurement, the overlay tests the complete experimental model. The off-condition $m = 0$ peaks show why a narrow Gaussian-like mosaic core is insufficient and why the Lorentzian-like tail is required, while the near-origin low-L feature emphasizes the importance of finite beam and wavelength phase space.

The PbI_2 section outlines how the same framework can be extended from ordered Bragg line-shape validation to diffuse scattering from stacking disorder. In that extension, the detector geometry, beam phase space, mosaicity, and projection procedure remain the same, but the reciprocal-space intensity is generated from a stacking-fault correlation model rather than from a perfectly periodic ordered stack. This separation is important: diffuse intensity should be interpreted as stacking information only after geometry and mosaicity have been constrained by the ordered-film workflow.

The practical refinement workflow is therefore part of the scientific argument. Instrumental parameters are determined first, sample alignment second, mosaic and profile parameters third, and ordered or diffuse intensity parameters last. This ordering reduces parameter compensation and makes the final data/model comparison interpretable. The resulting manuscript structure follows the same logic: show the detector features that require each model ingredient, explain the physical effect in reciprocal-space terms, and then use measured/calculated overlays as the validation.